STEG - Agni Pariksha

Category: Steganography | **Flag:** `PINAKA{gupt4_l1p1_4gn1_s4ty4m}`

The PNG

A 1400x1000 RGB PNG with six `iTXt` chunks. There's a 684-byte ZIP stuck on after `IEND`. The Sanskrit comment in the metadata points to the Panchavati forest, and `PANCHAVATI` turns out to be the ZIP password.

The ZIP

It's AES-encrypted, so plain `unzip` won't open it. `7z` works. Inside is `agni_kundal_mantra.txt`, Sanskrit instructions describing a spiral going outward from the fire pit. It gives the centre pixel, how much the spiral grows per turn, the channel (red), the bit (LSB) and a paired-alphabet cipher.

Reading the spiral

Clockwise Archimedean spiral from `(700, 500)`, growing `3.7` px per full turn, stepping `0.5` rad at a time. Coordinates get floored, pixels outside the image are skipped, and pixels already visited are skipped too. I read the red LSB of the first 800 pixels, MSB first. The first 4 bytes are a big-endian length (30), followed by:

```
CVANXN{thcg9_y6c6_9ta6_f9gl9z}
```

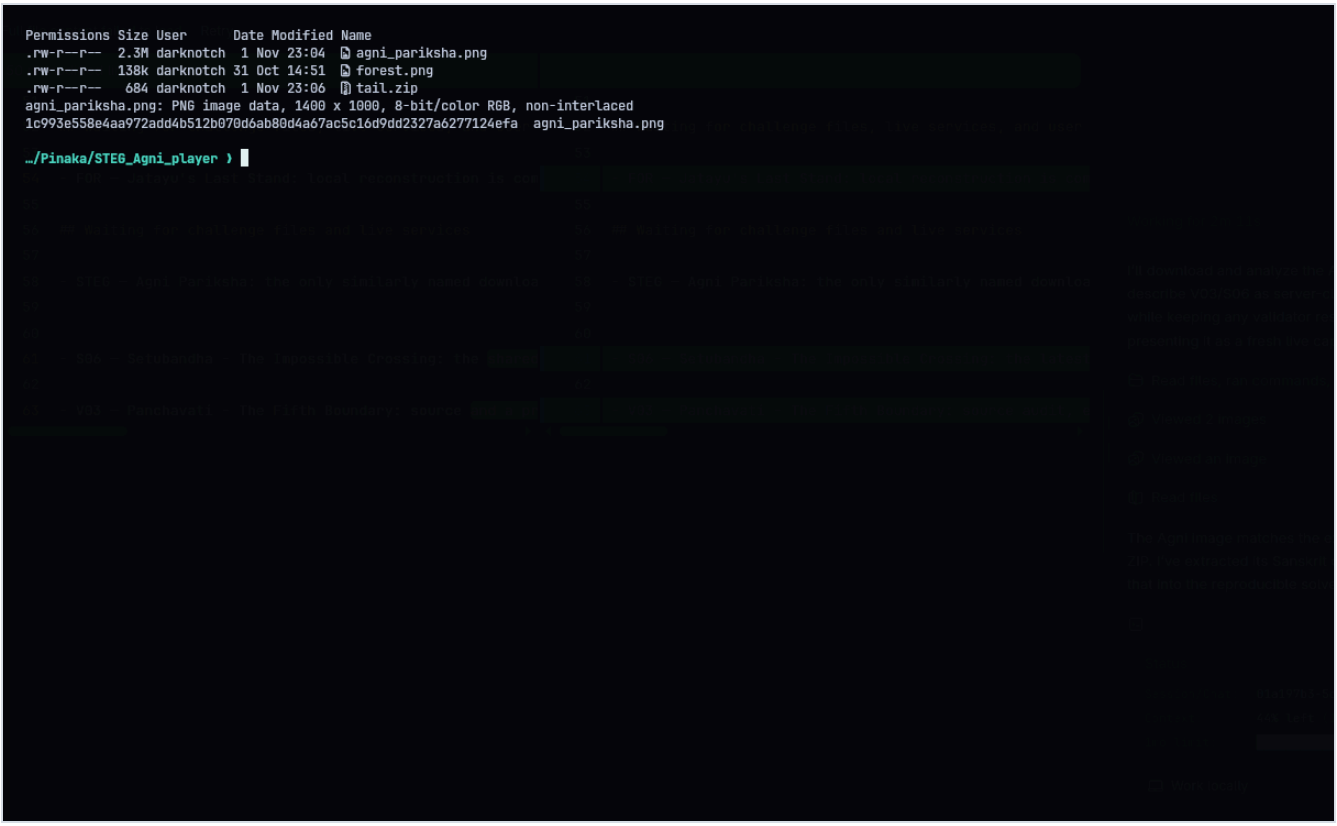
ROT13 on the letters and ROT5 on the digits:

```
PINAKA{gupt4_l1p1_4gn1_s4ty4m}
```

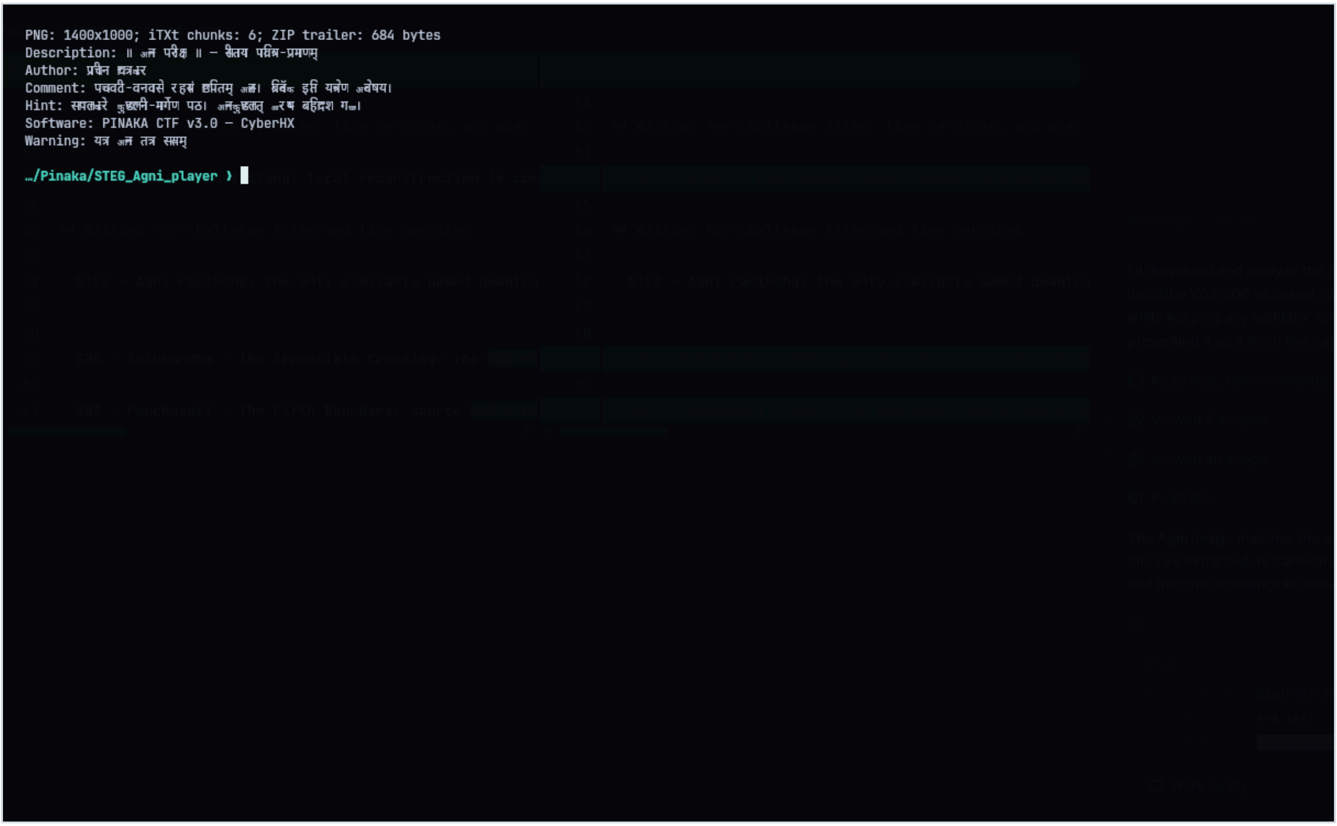
Reproduce

Needs the packages in `requirements.txt` and `7z` on the system. No online validator for this one.

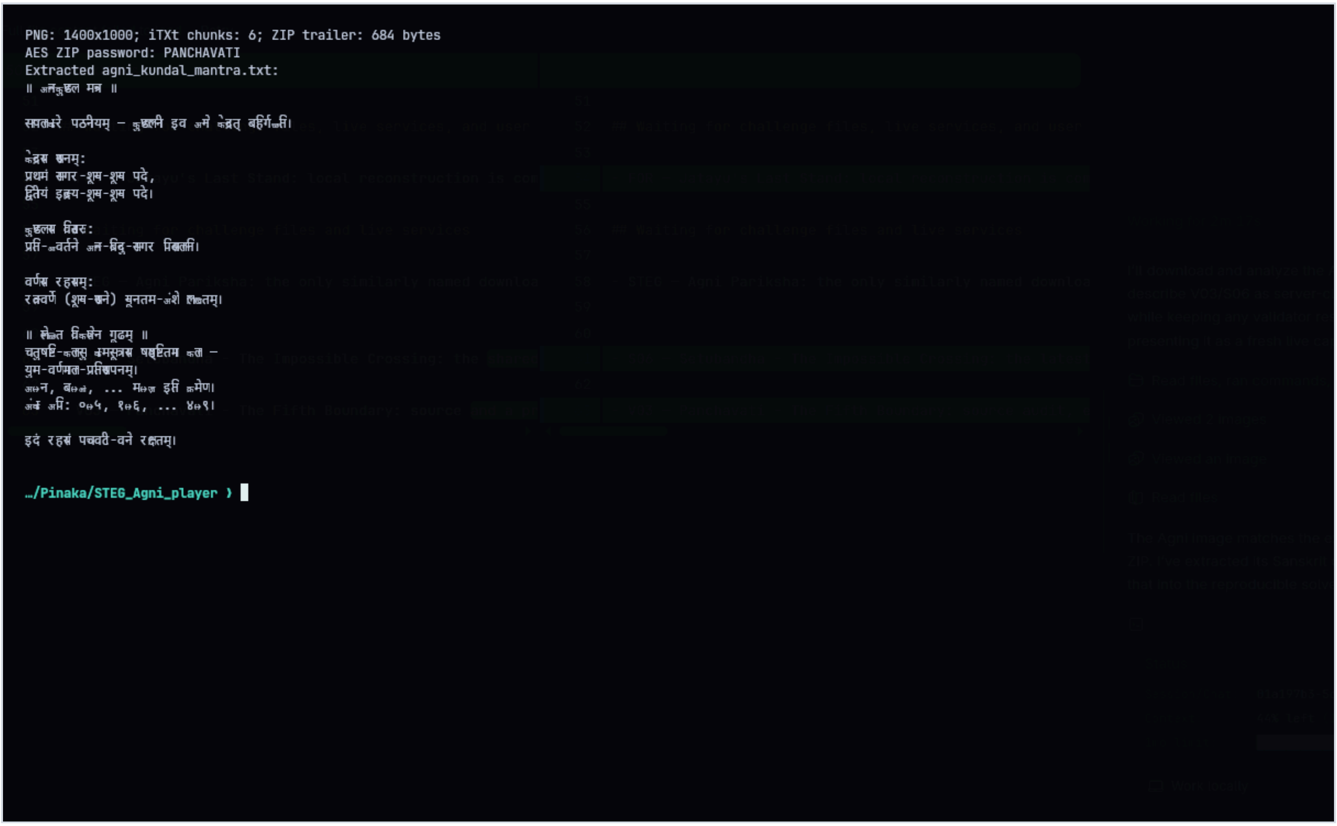
```
./repro.sh --stage metadata
./repro.sh --stage archive
./repro.sh --stage spiral
./repro.sh
```



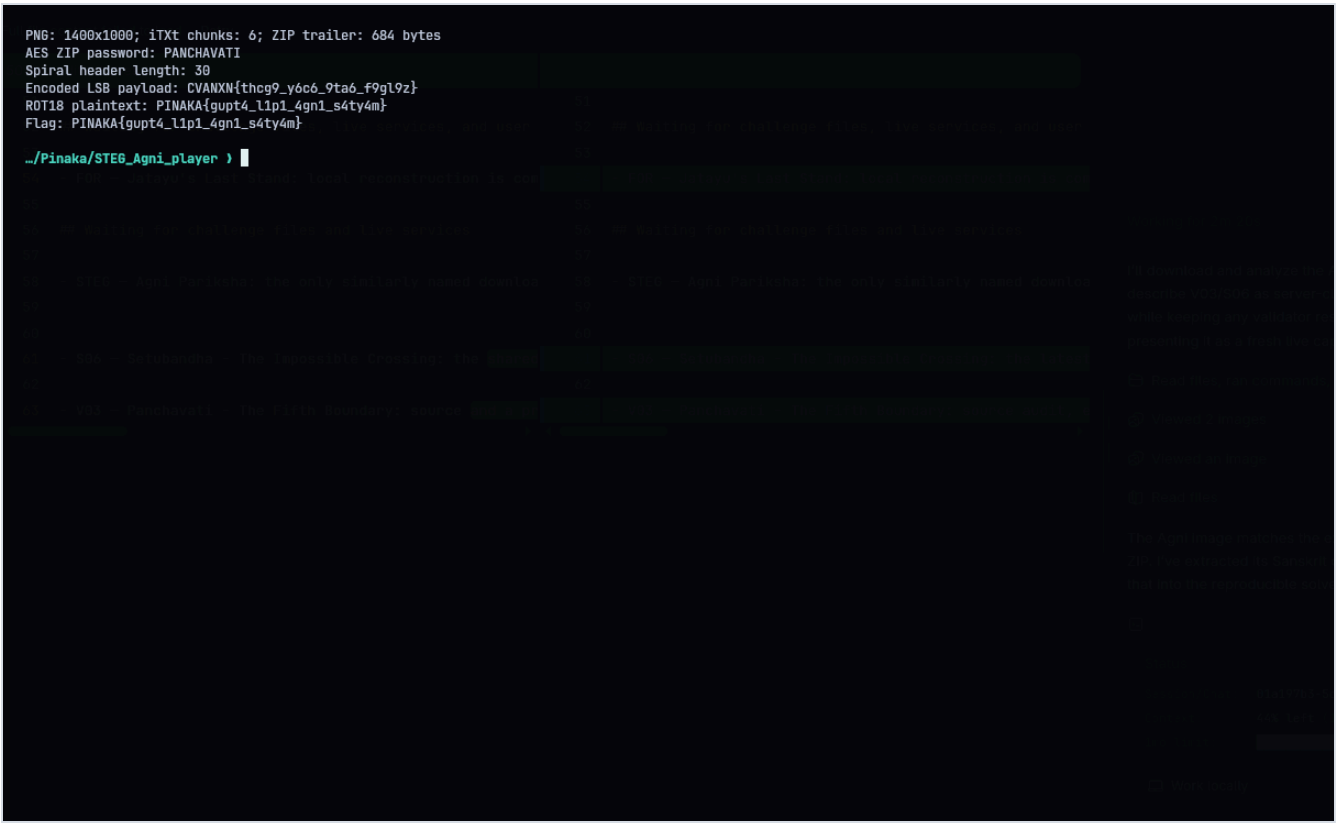
```
Permissions Size User      Date Modified Name
.rw-r--r--  2.3M darknotch  1 Nov 23:04  agni_pariksha.png
.rw-r--r--  138k darknotch 31 Oct 14:51  forest.png
.rw-r--r--   684 darknotch  1 Nov 23:06  tail.zip
agni_pariksha.png: PNG image data, 1400 x 1000, 8-bit/color RGB, non-interlaced
1c993e558e4aa972add4b512b070d6ab80d4a67ac5c16d9dd2327a6277124efa  agni_pariksha.png

…/Pinaka/STEG_Agni_player ❯ █
```



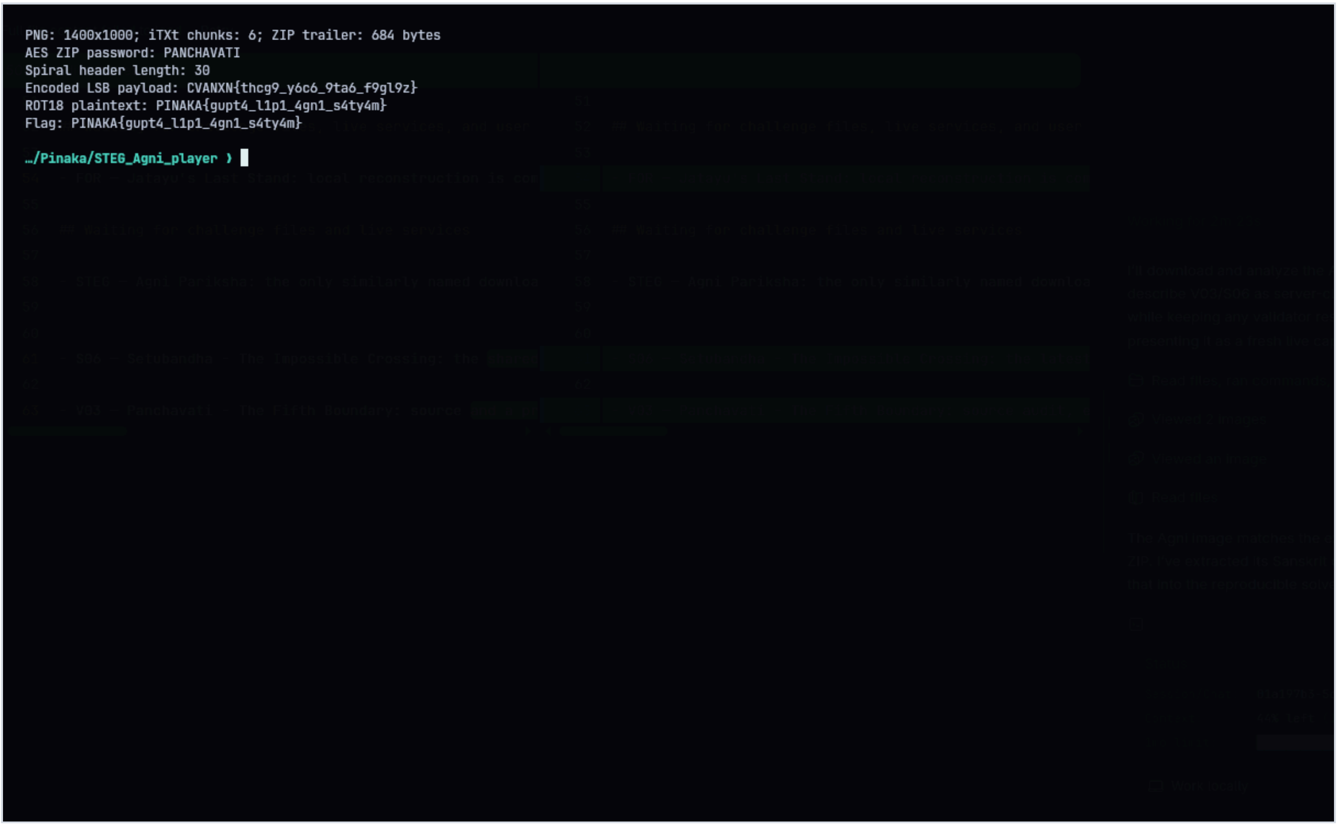
```
PNG: 1400x1000; iTXt chunks: 6; ZIP trailer: 684 bytes
Description: ॥ अग्नि परीक्षा ॥ — सीताया पवित्र-प्रमाणम्
Author: प्रचीन छत्रकर
Comment: पञ्चवटी-वनवासे रहस्यं छादितम् अग्नौ। बिन्वॉक इति यन्त्रेण अन्वेषय।
Hint: सप्तचक्रे कुण्डली-मार्गेण पठ। अग्निकुण्डात् आरभ्य बहिर्दिश गच्छ।
Software: PINAKA CTF v3.0 — CyberHX
Warning: यत्र अग्नि तत्र सत्यम्

…/Pinaka/STEG_Agni_player ❯ █
```



```
PNG: 1400x1000; iTXt chunks: 6; ZIP trailer: 684 bytes
AES ZIP password: PANCHAVATI
Extracted agni_kundal_mantra.txt:
॥ अग्निकुण्डल मन्त्र ॥

सर्पिलाकारे पठनीयम् – कुण्डलिनी इव अग्नेः केन्द्रात् बहिर्गच्छति।

केन्द्रस्य स्थानम्:
प्रथमं सागर-शून्य-शून्य पदे,
द्वितीयं इन्द्रिय-शून्य-शून्य पदे।

कुण्डलस्य विस्तारः:
प्रति-आवर्तने अन्त-बिन्दु-सागर विस्तरति।

वर्णस्य रहस्यम्:
रक्तवर्णे (शून्य-स्थाने) न्यूनतम-अंशे स्थितम्।

॥ संकेत विक्षेपेन गूढम् ॥
चतुष्षष्टि-कलासु क्रमसूत्रस्य षष्ठितम कला –
युग्म-वर्णमाला-प्रतिस्थापनम्।
अ↔न, ब↔ओ, ... म↔ज इति क्रमेण।
अङ्के अपि: ०↔५, १↔६, ... ४↔९।

इदं रहस्यं पञ्चवटी-वने रक्षितम्।


…/Pinaka/STEG_Agni_player ❯ █
```



```
PNG: 1400x1000; iTXt chunks: 6; ZIP trailer: 684 bytes
AES ZIP password: PANCHAVATI
Spiral header length: 30
Encoded LSB payload: CVANXN{thcg9_y6c6_9ta6_f9gl9z}
ROT18 plaintext: PINAKA{gupt4_l1p1_4gn1_s4ty4m}
Flag: PINAKA{gupt4_l1p1_4gn1_s4ty4m}

…/Pinaka/STEG_Agni_player ❯ █
```



```
PNG: 1400x1000; iTXt chunks: 6; ZIP trailer: 684 bytes
AES ZIP password: PANCHAVATI
Spiral header length: 30
Encoded LSB payload: CVANXN{thcg9_y6c6_9ta6_f9gl9z}
ROT18 plaintext: PINAKA{gupt4_l1p1_4gn1_s4ty4m}
Flag: PINAKA{gupt4_l1p1_4gn1_s4ty4m}

…/Pinaka/STEG_Agni_player ❯
```